

Shadi Mirmohammadi

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SUMMARY

Ph.D. candidate in Electrical & Computer Engineering specializing in **terahertz/mmWave device design, full-wave electromagnetic simulation**, and **semiconductor thin-film fabrication and characterization**. Designs reconfigurable photonic/RF devices in Ansys HFSS, fabricates and characterizes transparent-conductor films in the cleanroom, and automates analysis in Python. Directly relevant to **silicon photonics, optical interconnect, and mmWave hardware**.

EDUCATION

Ph.D., Electrical & Computer Engineering , University of Utah	Expected May 2028
M.Sc., Materials Chemistry , University of Utah	2023–2025
M.Sc., Nanochemistry , Iran University of Science & Technology	2017–2020

TECHNICAL SKILLS

- **EM Simulation & Device Design:** Ansys HFSS (full-wave 3D EM, Floquet ports, parametric sweeps, S-parameter extraction); COMSOL Multiphysics; resonator and metasurface design; impedance matching / critical coupling.
- **Fabrication & Process:** Magnetron sputtering (Denton Discovery 18) of ITO on Ga₂O₃/sapphire, maskless laser lithography (Heidelberg DWL66+), two-photon 3D lithography (Nanoscribe), thermal evaporation, RTP annealing, spin coating, high-vacuum & RF-plasma systems; cleanroom practice.
- **Optical & Electrical Characterization:** CW-THz photomixing spectroscopy (Toptica), THz time-domain spectroscopy, four-point probe, spectroscopic ellipsometry (WVASE; Cauchy, Tauc–Lorentz, Lorentz-oscillator), UV–Vis, photoluminescence, XPS, XRD, SEM/TEM, FIB/SIMS (training).
- **Programming & Analysis:** Python (NumPy, pandas, Matplotlib), MATLAB, L^AT_EX; simulation automation, data-analysis pipelines, numerical modeling, spectral fitting.

RESEARCH EXPERIENCE

Graduate Research Assistant — Sensale-Rodriguez / Blair Labs, University of Utah 2023–Present

- Designed and simulated **optically tunable THz/mmWave metasurface modulators** on Fe-doped β -Ga₂O₃ using full-wave 3D EM simulation (Ansys HFSS), achieving **~14 dB extinction ratio** and **~3 dB insertion loss** at ~300 GHz.
- Performed **S-parameter extraction and parametric optimization** across 13 conductivity states, identifying **critical-coupling** conditions yielding **~82% absorption** and **~99% reflection modulation depth** (H-SRR design; IRMMW-THz 2026).
- Developed **ITO thin-film sputter deposition recipes** (Denton Discovery 18) for transparent-electrode integration; validated films against reference **sheet-resistance** values by four-point probe and designed RTP anneal process (450°C, Ar).
- Performed **CW-THz transmission spectroscopy** (Toptica photomixing spectrometer) on ITO/sapphire films to extract THz-frequency sheet resistance; reconciling THz-extracted vs. DC four-point-probe conductivity.
- Built **Python data-analysis pipelines** (pandas, NumPy, Matplotlib) automating HFSS post-processing and measurement analysis; published open-source on GitHub.
- Characterized **excitation-selective Eu³⁺ emission switching** in GdScO₃ single crystals via photoluminescence spectroscopy; first-author manuscript in preparation for *Optical Materials Express*.
- Analyzed **UV photoresponsivity of GeO₂ photodetectors** (cross-group collaboration) with beam/detector-area corrections and calibrated power measurement.

Graduate Researcher — Materials Chemistry, University of Utah 2023–2025

- Synthesized and surface-engineered **semiconductor quantum dots** (CdSe, CdS, Si); developed **electrochemical sol-gel thin-film processing** and covalent surface-functionalization (hydrosilylation) routes; probed **spin-correlated radical pairs** via time-resolved EPR (Argonne National Lab).

SELECTED PUBLICATIONS & PRESENTATIONS

- *Optically Tunable THz Metasurface Absorber Based on Fe-Doped β -Ga₂O₃*, **IRMMW-THz 2026** (accepted), Salt Lake City, Oct 2026.
- *Excitation-Selective Eu³⁺ Emission in GdScO₃ Single Crystals*, manuscript in preparation for *Opt. Mater. Express*.
- *Si QD Sensitization of Anthracene Triplet via Time-Resolved EPR Spectroscopy*, submitted to *JACS*.
- *Increasing Structural Chirality of Metal Nanocrystals via Ligand Engineering*, *Small*, 2025.
- *Effects of Morphology and Size of Nanoscale Drug Carriers on Cellular Uptake*, *RSC Adv.*, 2023.
- Three additional papers (*Nanoscale Adv.* 2022; *Front. Chem. Sci. Eng.* 2020; *RSC Adv.* 2020) — full list on Scholar.

CONFERENCES & PROFESSIONAL DEVELOPMENT

- **ASCENT Summer School on Semiconductor Science & Technology**, Univ. of Michigan — selected in-person participant (UWBG semiconductors, thin-film growth, device fab), Aug 2026.
- **Pacificchem 2025**, Honolulu — Poster: *Spin-Correlated Radical Pairs in Si QD–Organic Hybrids* (Dec 2025).
- **Excited State Processes (ESP-2025)**, Santa Fe — Poster: *Engineering Spin States in Si QD–Organic Hybrids* (June 2025).
- **Rocky Mountain Conf. on Magnetic Resonance**, Snowbird, UT (Aug 2025).
- **ISPF-2 (2024)** — Poster: *Electronic Coupling Between Anthracene Isomers and CdSe Nanoplatelets* (July 2024).

HONORS & AWARDS

- **Best Poster Award** — ESP-2025 (June 2025) · **EDGES Graduate Fellowship** — Univ. of Utah (2024).